

- VLMs excel, but benchmarks remain task-narrow.
  - Exact-match scoring yields brittle false negatives.
  - Heterogeneous skills and biases hinder fairness.
- Language sensitivity skews cross-lingual scores.
  - MMBench: all-around, fine-grained, bilingual eval.
  - CircularEval + strict QC boost assessment fidelity.

## MMBench—A Robust Benchmark for Vision-Language Models

Comprehensive, bilingual benchmark to holistically evaluate VLM abilities

Yuan Liu et al.

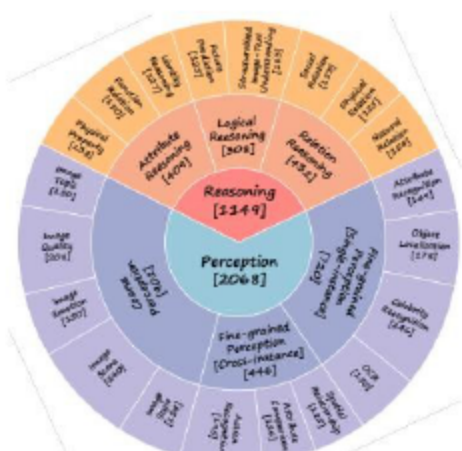
Shanghai AI Lab; CUHK; NTU; NUS; Zhejiang Univ.  
(2023)

### 3 Related Work: Datasets and VLMs

14 Yuan Liu et al.



## 5 Perception & Reasoning Ability Taxonomy



- Two pillars: Perception and Reasoning focus
  - Hierarchical splits into fine-grained abilities
  - Balanced sampling to cover diverse skills
- Comprehensive assessment across dimensions
  - Emulates human-like perception-reasoning flow
  - Supports targeted diagnostics and gap finding

## 6 Data Collection and Quality Control

- Multilingual item authoring in EN and ZH for broad coverage
- Curated images span daily scenes, documents, charts, and diagrams
- Manual collection from diverse sources for balanced coverage
  - Expert review and filtering; majority voting removes text-only items
  - Standardized MCQ formatting; inter-annotator checks ensure correctness
- Validation and leakage control ensure fair, unbiased evaluation

## 8 LLM-Involved Choice Extraction Workflow

- Extract multi-choice options from VLM outputs
  - Robust prompts handle diverse answer formats
  - Normalize bilingual responses in EN and ZH
- Mitigate formatting variance across models
  - Increase evaluation accuracy over rule-based
  - Accommodate varied instruction-following

## Sec 2 Evaluation Strategy

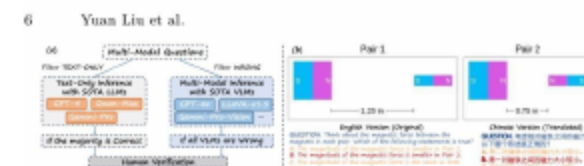


Fig. 3: The construction of MMBench. (a). The quality control strategies adopted in MMBench. (b). An illustration of questions in MMBench-EN.

### 3 CircularEval & LLM-Human Align.

Table 1: Statistics of IF capabilities of VLMs. We report the heuristic matching success rate of VLMs, and the accuracy before and after LLM-based choice extraction. In 'X+Y', X denotes the matching-based accuracy, Y indicates the gain of using LLM as the choice extractor.

| Model Name          | Match Rate | DEV Acc    | Model Name       | Match Rate | DEV Acc    |
|---------------------|------------|------------|------------------|------------|------------|
| MiniGPT4v           | 65.7       | 47.9 +17.8 | MiniGPT4v-1.5B   | 54.8       | 52.1 +12.7 |
| InternVL2v1.5       | 90.6       | 52.1 +38.5 | InternVL2v1.5-7B | 90.7       | 55.4 +35.3 |
| DEFACTO-2B-Instruct | 96.6       | 58.4 +38.2 | Qwen-VL-Chat     | 93.8       | 73.3 +20.5 |
| MiniGPT4v           | 80.4       | 79.8 +0.6  | VisualGLM-4B     | 63.8       | 36.9 +26.9 |
| GPT-4v              | 90.8       | 88.3 +2.5  | Qwen2-VL-72B     | 97.5       | 81.9 +15.6 |
| Qwen-VL-Plus        | 77.4       | 81.3 +4.9  | Qwen-VL-Max      | 90.8       | 82.8 +8.0  |

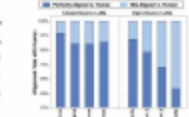


Fig. 5: Alignment rates between human and different LLMs. 'chatgpt' is 'gpt-3.5-turbo'. Open-source LLMs are 'chat' variants.

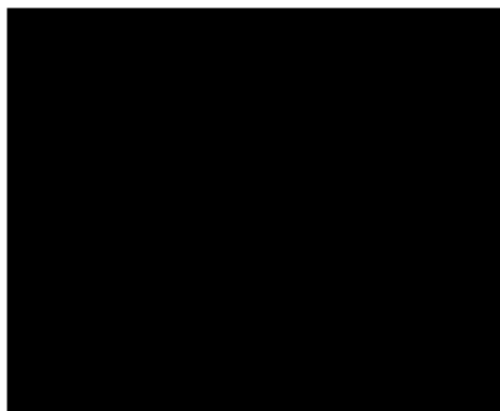


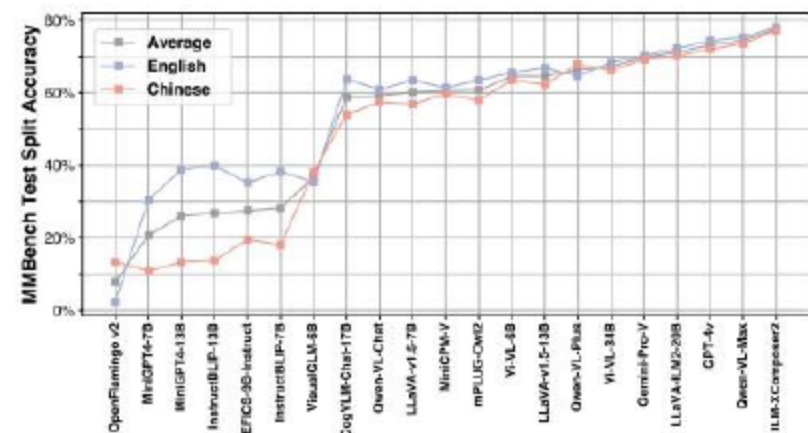
Fig. 6: CircularEval strategy. In CircularEval, a problem is tested multiple times with circularly shifted choices and the VLM needs to succeed in all testing passes. In this

- Present each question multiple times with permuted options.
  - Correct choice must stay invariant across permutations.
  - Mitigates positional/chance bias in MC-style evaluation.
- Report match rate, DEV accuracy, and LLM-human alignment.

Sec 10

## Results: Performance & Language

### 11 Performance & Language Sensitivity



- Proprietary models lead; GPT-4v excels overall
  - Open-source like InternLM-XComposer gain via tuning
  - CircularEval exposes gaps in tuning/alignment
- English vs Chinese accuracy diverges by model
  - Perception vs reasoning drive task differences
  - Balanced corpora and cross-lingual benchmarks matter

### 12 Conclusion: Key Findings and Impact

- Fine-grained, bilingual, taxonomy-driven VLM benchmark.
- Robust evaluation via LLM extraction and CircularEval.
- Comprehensive MCQ framework surfaces capability gaps.
  - Reveals language sensitivity across models and tasks.
  - Sets a new, objective standard for VLM evaluation.
- Guides future VLM research and cross-lingual generalization.